

Release Plan: CircleHome

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High Level Goals

- *Household Task Management*: Users can create, categorize, assign, and complete household and care tasks from a unified feed.
 - Users can also be a part of multiple homes (like college students able to have one for their family home and their college home)
 - Members of a household can add recurring or one-time tasks, label them as household or care tasks, and mark them complete. The feed updates in real time across all household members.
 - These tasks are also ranked from 1 to 5 for level of difficulty to enable the Smart Rotation Engine and delegated to people through points
 - Later update: members for can offer to swap tasks with each other and both parties must accept before assignment is confirmed
- *Smart Rotation Engine*: Tasks are auto-assigned based on each member's configured weekly capacity.
 - The algorithm distributes tasks fairly using a weighted round-robin model. When a member's capacity is reduced (e.g., they are handling heavy care duties) aka already have a high amount of points for their capacity, the engine automatically offloads household tasks to others.
- *Care Diary and Health Log*: Caregivers can log structured notes after care visits that build a longitudinal health record. (Guest access?)
 - Notes cover mood, appetite, symptoms, and free text.
 - Later update: An AI-generated weekly summary condenses the log into plain English and can be exported as a PDF for physician review.
- *Escalation and Notifications*: Overdue care tasks automatically notify a backup household member.
 - Push notifications and optional SMS alerts ensure that critical care tasks never go unacknowledged. The care recipient can also be notified directly as a last resort.
- *Household Health Score*: A shared balance metric shows the overall fairness of task distribution without requiring confrontation.
 - The score is visible to all members and updates weekly, encouraging contribution through social motivation rather than nagging.
 - Couple ideas for the visualization: a circle, leaderboard

Sprint 1: Foundation + Setup

- *Goal:* Sprint 1 focuses on getting the core infrastructure in place so that every subsequent sprint can build on a stable foundation. The goal is to have user authentication, household creation and joining, and basic task creation all working end-to-end by the close of the sprint. Nothing should be polished at this stage but everything should be functional.
- Total Story Points: 22

Priority	User story	Points	Type/ Member
1	(1.1) As a new user, I want to create an account with an email and password so that I can access my household securely.	3	Backend
2	(1.2) As a new user, I want to create a household or join one using an invite code so that I can coordinate with the people I live with.	3	Backend
3	(1.3) As a household member, I want to create a task with a title, category, due date, and assigned member so that responsibilities are clearly defined.	3	
4	(1.4) As a household member, I want to view all household tasks in a shared feed sorted by due date so that I always know what needs to be done.	3	Frontend
5	(1.5) As a household member, I want to mark a task as complete so that the feed reflects current status and the assignee gets credit.	2	Frontend
6	(1.6) As a developer, I want a Flutter project scaffold with a Firebase backend, authentication, and a Firestore schema for households, users, and tasks so that all features have a stable foundation.	5	Pravin (Backend)
7	(1.7) As a household member, I want the task feed to update in real time when a teammate adds or completes a task so that I do not have to refresh the app.	3	

Sprint 2: Core Task System

- *Goal:* Sprint 2 builds out the core task functionality on top of Sprint 1's foundation. The goal is to have task creation, assignment, and completion fully wired between the Flutter UI and Firestore, with basic filtering so members can toggle between their tasks and all household tasks.
- Total Story Points: 22

Priority	User story	Points	Type/ Member
1	(2.1) As a household member, I want to create a task so that I can add responsibilities to the shared feed.	3	Backend
2	(2.2) As a household member, I want to assign a task to a specific household member so that responsibilities are clearly delegated.	3	Backend
3	(2.3) As a household member, I want to view both my assigned tasks and all household tasks so that I have full visibility into what needs to be done.	3	Frontend
4	(2.4) As a household member, I want to mark a task as complete so that the feed reflects current status.	2	Frontend
5	(2.5) As a household member, I want my tasks to be saved and synced across devices so that I can access them anytime from anywhere.	4	Backend
6	(2.6) As a household member, I want my actions (creating, completing, viewing tasks) to update instantly so that I always see the most current information.	4	Backend
7	(2.7) As a household member, I want to filter and organize tasks by status (e.g., completed, incomplete, overdue) so that I can better manage and prioritize my responsibilities.	3	Backend

Sprint 3: Accountability + Care + Notifications

- *Goal:* Sprint 3 introduces the care diary, a notification system for overdue or upcoming tasks, and a household activity dashboard. By the end of this sprint, caregivers should be able to log notes and all members should receive push alerts for relevant tasks.
- Total Story Points: 23

Priority	User story	Points	Type/ Member
1	(3.1) As a household member, I want to receive push notifications for upcoming or overdue tasks so that nothing slips through the cracks.	4	Backend
2	(3.2) As a caregiver, I want to log basic caregiving notes (mood, appetite, symptoms, free text) so that I can maintain a running health record.	3	Backend
3	(3.3) As a household member, I want to view past care notes so that I can track trends in the care recipient's wellbeing.	3	Frontend
4	(3.4) As a household member, I want to see a recent activity feed on the dashboard so that I can stay informed about what my household has been doing.	3	Frontend
5	(3.5) As a developer, I want to set up Firebase Cloud Messaging so that the notification system has a reliable delivery infrastructure.	4	Backend
6	(3.6) As a developer, I want to design and implement the care notes database schema so that structured notes can be stored and queried efficiently.	3	Backend
7	(3.7) As a developer, I want to build a dashboard view combining tasks and recent activity so that members have a unified home screen.	3	Frontend

Sprint 4: Leaderboard + Polish + MVP Completion

- *Goal:* Sprint 4 wraps up the MVP. The leaderboard and per-member contribution view are implemented, the UI is polished and consistent across all screens, and the app is stable enough for a live demo. This sprint is lighter on new features to leave room for bug fixes, testing, and demo prep.
- Total Story Points: 20

Priority	User story	Points	Type/ Member
1	(4.1) As a household member, I want to see a task completion leaderboard so that I can track how much each person is contributing.	4	Frontend
2	(4.2) As a household member, I want to view each member's task contributions so that fairness is transparent without requiring confrontation.	3	Frontend
3	(4.3) As a user, I want a clean and consistent UI across all pages so that the app feels polished and professional.	3	Frontend
4	(4.4) As a developer, I want to design the leaderboard scoring logic (weighted by task difficulty points 1-5) so that the rankings reflect actual effort fairly.	3	Backend
5	(4.5) As a developer, I want to perform UI polish across all pages so that visual inconsistencies are resolved before the demo.	2	Frontend
6	(4.6) As a developer, I want to perform bug fixes and run end-to-end testing so that the app is stable for the final release.	3	Backend
7	(4.7) As a developer, I want to complete final integration and prepare a demo build so that the team is ready to present CircleHome.	2	All

Sanity Check

Each Sprint should have a maximum of 25 story points for 100 points total. Plan accordingly for midterms/finals as well.

Product Backlog

Bleh bleh bleh → tasks that we are unable to complete